

MANTA, Ecuador

WMO: 841170

Lat: 0.950S

Lon: 80.683W

Elev: 15

StdP: 101.15

Time Zone: -5.00 (W05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	19.8	20.1	16.8	12.0	22.3	17.0	12.2	22.2	9.0	25.6	8.3	25.3	3.1	210	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.2	30.9	24.6	30.1	24.6	29.8	24.6	26.3	28.9	26.0	28.6	25.6	28.3	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	27.9	25.1	20.2	27.7	24.8	19.9	27.5	82.1	29.4	80.5	28.7	78.8	28.4	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	7.2	6.5	19.1	33.0	1.0	0.8	18.4	33.6	17.8	34.1	17.2	34.5	16.5	35.1		
			17.4	26.9	1.5	1.1	16.3	27.7	15.4	28.4	14.5	29.0	13.4	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.3	26.1	26.1	26.3	26.1	26.0	25.1	24.7	24.4	24.4	24.3	24.6	25.5
	DBStd	1.44	1.15	0.91	0.85	1.11	1.23	1.43	1.43	1.31	1.31	1.15	1.26	1.26
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5581	499	452	506	483	497	453	454	446	431	442	439	479
	CDD18.3	2539	241	218	248	233	238	203	196	187	181	184	189	221
	CDH23.3	15439	1823	1743	2071	1844	1743	1116	897	767	703	661	795	1276
	CDH26.7	2663	307	296	398	356	367	186	138	111	98	79	118	209
Wind	WSAvg	3.4	3.0	2.3	2.3	2.5	3.1	3.6	3.8	3.9	4.0	4.1	4.1	3.9
Precipitation	PrecAvg	484	78	129	116	68	22	8	2	3	3	3	9	18
	PrecMax	1672	374	315	355	229	279	109	22	40	33	36	195	267
	PrecMin	136	11	47	8	0	0	0	0	0	0	0	0	0
	PrecStd	369	82	77	95	70	61	23	5	8	6	7	39	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	30.2	30.8	31.1	31.2	30.9	30.2	30.7	30.1	30.1	30.1	31.2
		MCWB	23.9	25.3	25.8	25.5	24.8	24.6	24.0	23.7	23.8	23.7	23.5	23.9
	2%	DB	29.9	29.8	30.0	30.1	30.1	29.8	29.1	29.0	29.0	28.3	29.1	29.9
		MCWB	24.2	25.2	25.4	25.0	24.8	24.5	23.7	23.4	23.5	23.2	23.4	23.9
	5%	DB	29.0	29.1	29.2	29.2	29.2	28.8	28.1	27.9	27.8	27.2	28.1	28.8
		MCWB	24.2	24.9	25.1	24.6	24.5	24.1	23.4	23.0	23.0	22.8	23.1	23.5
	10%	DB	28.2	28.2	28.8	28.6	28.8	27.8	27.1	26.9	26.8	26.3	27.1	27.9
		MCWB	23.9	24.5	24.9	24.4	24.4	23.5	23.0	22.6	22.7	22.4	22.7	23.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.7	27.0	26.5	26.6	26.0	25.2	24.7	24.9	24.7	25.5	26.0
		MCDB	28.9	29.0	29.4	29.4	29.7	29.3	28.8	28.6	28.8	28.5	28.3	29.4
	2%	WB	25.6	26.1	26.3	25.9	25.8	25.1	24.4	23.8	24.0	23.8	24.6	25.1
		MCDB	28.3	28.4	28.7	28.7	29.0	28.4	27.9	27.8	27.6	27.5	27.5	28.0
	5%	WB	25.1	25.7	25.9	25.4	25.2	24.4	23.7	23.3	23.4	23.3	23.8	24.4
		MCDB	27.6	27.9	28.2	28.2	28.3	27.5	27.0	26.8	26.7	26.6	26.8	27.4
	10%	WB	24.6	25.2	25.5	24.9	24.7	23.8	23.3	22.7	22.7	22.7	23.1	23.6
		MCDB	27.2	27.5	27.8	27.7	27.7	26.7	26.4	25.9	25.8	25.6	26.2	26.8
Mean Daily Temperature Range	5% DB	MDBR	5.1	4.9	5.2	5.4	5.6	5.5	5.5	5.8	5.6	5.5	5.8	5.5
		MCDBR	5.9	5.6	5.8	6.0	6.4	6.5	6.6	6.9	6.9	6.5	7.0	6.7
	5% WB	MCWBR	2.7	2.8	3.0	2.9	3.0	3.1	3.1	3.1	3.1	3.2	3.2	3.0
		MCWBR	5.4	5.2	5.4	5.6	6.0	6.1	6.0	6.3	6.2	6.2	6.4	6.1
Clear-Sky Solar Irradiance	taub	taub	0.439	0.459	0.467	0.479	0.480	0.471	0.473	0.499	0.525	0.485	0.484	0.474
		taud	2.396	2.334	2.311	2.271	2.273	2.303	2.304	2.210	2.120	2.248	2.241	2.279
	Ebn at Noon	Ebn at Noon	894	878	864	833	807	801	807	804	803	845	846	856
		Edh at Noon	125	134	137	138	133	126	128	145	163	144	145	139
All-Sky Solar Radiation	RadAvg	RadAvg	4.50	4.65	5.09	5.18	4.47	3.67	3.51	3.69	3.87	3.89	4.16	4.39
	RadStd	RadStd	0.40	0.36	0.37	0.37	0.37	0.45	0.32	0.33	0.37	0.37	0.36	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page